Noncommutative Davis inequalities and applications

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Davis inequalities

- Let $(\Omega, \mathcal{F}, \mathbb{P}; \{\mathcal{F}_n\}_{n\geq 1})$ be a probability space associated with an increasing filtration $\{\mathcal{F}_n\}_{n\geq 1}$.
- Let $\{\mathbb{E}_k\}_{k\geq 1}$ be the conditional expectation with respect to $\{\mathcal{F}_n\}_{n\geq 1}$.
- For a scalar martingale $f_n = \mathbb{E}_n f$, let $d_n f = \mathbb{E}_n(f) - \mathbb{E}_{n-1}(f)$, and

$$f^* = \sup_{n\geq 1} |f_n|, \qquad S(f) = \left(\sum_{n\geq 1} |d_n f|^2 \right)^{1/2}.$$

The Burkholder–Gundy inequalities compare maximal and square functions for $1 < p < \infty$.

Theorem (Burkholder, 1966)

There exist $c_p, C_p > 0$ such that

$$c_p \|S(f)\|_p \leq \|f^*\|_p \leq C_p \|S(f)\|_p, \quad 1 < p < \infty.$$

Davis inequalities

Definition

A filtration is regular if for each n

$$\mathbb{E}_n(f) \leq c_{\mathrm{reg}}\, \mathbb{E}_{n-1}(f), \qquad f \geq 0,$$

with one constant c_{reg} independent of a and n.

Theorem (Burkholder–Gundy, 1970, Acta Math.)

Under the regular condition, there exists $c_p, C_p > 0$ such that

$$c_p \left\|S(f)\right\|_p \leq \left\|f^*\right\|_p \leq C_p \left\|S(f)\right\|_p, \quad 0 < p \leq 1.$$

Theorem (Davis, 1970)

For general martingale, there is a universal constant $C > 0$ such that

$$C^{-1} \left\|S(f)\right\|_1 \leq \left\|f^*\right\|_1 \leq C \left\|S(f)\right\|_1.$$

Here we study the Davis inequality for noncommutative martingales

Noncommutative martingale theory

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- N. Randrianantoanina (2002). *Non-commutative martingale transforms*. J. Funct. Anal. **194**(1), 181–212.
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Noncommutative martingales

• Let $(\mathcal{M}, \tau)$ be a finite von Neumann algebra with a normal faithful normalized trace. For $0 < p < \infty$,

$$\|x\|_p = \tau(|x|^p)^{1/p}.$$

• Consider an increasing, weak*-dense filtration

$$\mathcal{M}_1 \subset \mathcal{M}_2 \subset \cdots \subset \mathcal{M}, \qquad \mathcal{E}_n : \mathcal{M} \longrightarrow \mathcal{M}_n,$$

where each $\mathcal{E}_n$ preserves the trace.

• For convenience, we here only consider the noncommutative martingale $x := (\mathcal{E}_n(x_{n+1}))_n = (x_n)_{n\geq 1}$. Its differences are

$$d_1 x = x_1, \qquad d_n x = x_n - x_{n-1} \quad (n \geq 2).$$

Nc Burkholder-Gundy

Noncommutativity produces two square functions:

$$S_c(x) = \left(\sum_{n \geq 1} |d_n x|^2 \right)^{1/2}, \qquad S_r(x) = \left(\sum_{n \geq 1} |d_n x^*|^2 \right)^{1/2}.$$

Let

$$\|x\|_{H_p^c} = \|S_c(x)\|_p, \qquad \|x\|_{H_p^r} = \|S_r(x)\|_p.$$

Theorem (Pisier and Xu, 1997)

For $1 < p \leq 2$,

$$L_p(\mathcal{M}) \sim H_p(\mathcal{M}) := H_p^c(\mathcal{M}) + H_p^r(\mathcal{M}),$$

$$\|x\|_{H_p} = \inf_{x=y+z} \left\{ \|y\|_{H_p^c} + \|z\|_{H_p^r} \right\},$$

where y and z are martingales.
For $2 \leq p < \infty$,

$$L_p(\mathcal{M}) \sim H_p(\mathcal{M}) = H_p^c(\mathcal{M}) \cap H_p^r(\mathcal{M}),$$

$$\|x\|_{H_p} = \max \left\{ \|y\|_{H_p^c}, \|z\|_{H_p^r} \right\},$$

Noncommutative maximal spaces

The noncommutative maximal norm uses factorizations in place of a pointwise supremum:

$$x_n = a\, v_n\, b, \qquad a, b \in L_{2p}(\mathcal{M}), \qquad (v_n) \subset \mathcal{M}.$$

$$\|(x_n)\|_{L_p(\mathcal{M},\ell_\infty)} = \inf_{x_n = a v_n b} \left\{ \|a\|_{2p} \sup_n \|v_n\|_\infty \|b\|_{2p} \right\}.$$

Positive sequences

If every $x_n \geq 0$, then

$$\|(x_n)\|_{L_p(\mathcal{M},\ell_\infty)} = \inf \left\{ \|A\|_p : A \in L_p(\mathcal{M})_+,\ x_n \leq A \text{ for all } n \right\}.$$

Thus a positive majorant A replaces the classical maximal function.

The endpoint obstruction

Junge's noncommutative Doob inequality gives

$$\|(\mathcal{E}_n x)_n\|_{L_p(\mathcal{M},\ell_\infty)} \leq C_p \|x\|_p, \qquad 1 < p \leq \infty.$$

Junge and Xu identified the optimal growth $C_p \asymp (p-1)^{-2}$ as $p \downarrow 1$.

Failure of one direction at $p = 1$

Junge and Xu (2005) showed that no universal C can satisfy

$$\|(\mathcal{E}_n x)_n\|_{L_1(\mathcal{M},\ell_\infty)} \leq C \|x\|_{H_1(\mathcal{M})}$$

for all noncommutative martingales.

Hong, Junge and Parcet (2016) obtained an asymmetric maximal substitute.

Conditioned square functions

For a finite L_2-martingale, define

$$s_c(x) = \left(|d_1 x|^2 + \sum_{n \geq 2} \mathcal{E}_{n-1}(|d_n x|^2) \right)^{1/2}.$$

The conditioned and diagonal Hardy norms are

$$\|x\|_{h_p^c} = \|s_c(x)\|_p, \qquad \|x\|_{h_p^d} = \left(\sum_{n \geq 1} \|d_n x\|_p^p \right)^{1/p}.$$

Perrin (2009); Junge–Mei (2010)

For $x \in H_1^c(\mathcal{M})$,

$$\|x\|_{H_1^c} \simeq \inf_{x=y+z} \left\{ \|s_c(y)\|_1 + \sum_{n \geq 1} \|d_n z\|_1 \right\},$$

Here y, z are martingales, and the filtration is arbitrary.

M. Perrin, J. Lond. Math. Soc. **80** (2009), 627–648.

M. Junge and T. Mei, Amer. J. Math. **132** (2010), 611–680.

Junge–Perrin's diagonal refinement

For $1 \le p < 2$, let $1/p = 1/2 + 1/r$. Define the column diagonal norm

$$\|(a_n)\|_{L_p(\mathcal{M};\ell_1^c)} = \inf_{a_n = b_n c_n} \left(\sum_n \|b_n\|_2^2 \right)^{1/2} \left\| \left(\sum_n |c_n|^2 \right)^{1/2} \right\|_r,$$

and set $\|z\|_{h_p^{1c}} = \|(d_n z)_n\|_{L_p(\mathcal{M};\ell_1^c)}$.

Junge–Perrin (2014), Theorem 6.1.6

For $1 \le p < 2$, with equivalent norms,

$$H_p^c(\mathcal{M}) = h_p^c(\mathcal{M}) + h_p^{1c}(\mathcal{M}).$$

Moreover, $h_p^{1c}(\mathcal{M}) \subset h_p^d(\mathcal{M})$ contractively.

At $p = 1$, $\|(a_n)\|_{L_1(\mathcal{M};\ell_1^c)} = \sum_n \|a_n\|_1$. For $1 < p < 2$, this refines the diagonal part and supports continuous filtrations.

M. Junge and M. Perrin, Theory of $\mathcal{H}_p$-spaces for continuous filtrations in von Neumann algebras, Astérisque **362** (2014), §6.1.

Constructive Davis decompositions

Randrianantoanina–Wu–Xu (2019), Corollary 3.4

Let $1 \le p < 2 \le q < \infty$ and $x \in H_p^c(\mathcal{M}) \cap H_q^c(\mathcal{M})$. There exist martingales y, z with $x = y + z$ such that

$$\|y\|_{h_p^c} + \|z\|_{h_p^d} \le 2^{5/2} \|x\|_{H_p^c},$$

and the same decomposition satisfies

$$\|y\|_{h_q^c} + \|z\|_{h_q^d} \le Cq \|x\|_{H_q^c},$$

where C is an absolute constant.

At $p = 1$, Corollary 3.6 also gives a decomposition with the endpoint norm bound and the predictable estimate

$$|d_n y|^2 \le 2S_{c,n-1}(x)^2, \qquad S_{c,n-1}(x)^2 = \sum_{k<n} |d_k x|^2.$$

N. Randrianantoanina, Lian Wu and Q. Xu, Noncommutative Davis type decompositions and applications, J. Lond. Math. Soc. **99** (2019), 97–126.

Junge and Xu's question

The converse direction of the endpoint inequality is the central problem.

Problem 16, Junge and Xu (2005)

Does there exist a universal constant $C > 0$ such that

$$\|x\|_{H_1(\mathcal{M})} \leq C \, \|(\mathcal{E}_n x)_n\|_{L_1(\mathcal{M},\ell_\infty)}$$

for every noncommutative martingale?

The scalar proof uses pointwise comparisons and stopping times. In a von Neumann algebra, one must work with projections and column–row decompositions.

Our result answers this question for regular filtrations.

Regular filtrations

Definition

A filtration is regular if successive conditional expectations satisfy

$$\mathcal{E}_n(a) \leq c_{\mathrm{reg}}\, \mathcal{E}_{n-1}(a), \qquad a \geq 0,$$

with one constant c_{reg} independent of a and n.

Basic example is the operator-valued dyadic filtration

$$\mathcal{M}_n = L_\infty(\mathcal{D}_n) \bar{\otimes} \mathcal{N}, \qquad \mathcal{E}_n = \mathbb{E}_n \otimes \mathrm{id}_{\mathcal{N}}.$$

Here $\mathcal{D}_n$ is generated by intervals of length 2^{-n}, and

$$c_{\mathrm{reg}} = 2.$$

Main result

Theorem 1.2 (Jiao–Luo–Sukochev–Zhou)

Let $(\mathcal{M}, \tau)$ be a noncommutative probability space with a regular filtration $(\mathcal{M}_n)_{n\geq 1}$. For every martingale $x = (x_n)_{n\geq 1}$ and every $0 < p \leq 1$,

$$\boxed{\|x\|_{H_p(\mathcal{M})} \leq C_{p,\mathrm{reg}} \, \|(x_n)_n\|_{L_p(\mathcal{M},\ell_\infty)}.}$$

The constant depends only on p and the regularity constant.

At $p = 1$, this gives an affirmative answer to Junge and Xu's question under regularity.

Remarks on the main theorem

- The estimate holds throughout $0 < p \leq 1$, including the quasi-Banach range $0 < p < 1$.
- The regularity assumption is essential to the present argument. The manuscript leaves the case of arbitrary filtrations open.
- The same argument extends to semifinite von Neumann algebras equipped with a regular filtration.
- Applications include Mei's operator-valued Hardy spaces and a dyadic Carleson decomposition for noncommutative BMO.

Ingredients of the proof

The argument combines estimates for stopping projections with a column–row decomposition.

1. Cuculescu projections describe the part of a positive martingale above a given level.
2. The Fuglede–Kadison determinant and Carleman's inequality control the sum of projection tails by a positive majorant.
3. Triangular truncations split the martingale into column and row pieces. A regularity estimate controls their square function distributions.

Summing the resulting dyadic estimates yields the H_p bound.

Cuculescu projections

Let $x_n = \mathcal{E}_n(x)$ be a positive martingale. For $\lambda > 0$, set $q_0^{(\lambda)} = 1$ and

$$q_n^{(\lambda)} = q_{n-1}^{(\lambda)} \chi_{[0,\lambda]}\left(q_{n-1}^{(\lambda)} x_n q_{n-1}^{(\lambda)}\right).$$

The projections decrease in n, belong to $\mathcal{M}_n$, and satisfy

$$q_n^{(\lambda)} x_n q_n^{(\lambda)} \leq \lambda q_n^{(\lambda)}.$$

Writing $q^{(\lambda)} = \bigwedge_n q_n^{(\lambda)}$, we have

$$\lambda \tau(1 - q^{(\lambda)}) \leq \tau\left((1 - q^{(\lambda)})x\right) \leq \|x\|_1 .$$

The quantity $\tau(1 - q^{(\lambda)})$ plays the role of a maximal distribution function.

Fuglede–Kadison determinant

For a positive invertible $b \in e\mathcal{M}e$, use the restricted, unnormalized trace:

$$\Delta_e(b) = \exp\big(\tau(\log_e b)\big).$$

Lemma 3.4 (Corner estimate)

Let $e = p + q$, where p and q are orthogonal projections. If $b \in (e\mathcal{M}e)_+$ is invertible and $b \geq \lambda p$, then

$$\Delta_e(b) \geq \lambda^{\tau(p)} \Delta_q(qbq).$$

For $b = \begin{pmatrix} a & c \\ c^* & d \end{pmatrix}$, the Schur complement satisfies $s = a - cd^{-1}c^* \geq \lambda p$.
Factorization gives

$$\Delta_e(b) = \Delta_p(s)\Delta_q(d).$$

A distribution estimate from a majorant

Suppose $0 \le x_n \le a$ for every n, and define

$$A(t) = \exp\left(\frac{1}{t}\int_0^t \log \mu(s,a)\,ds\right), \qquad t > 0,$$

where $\mu(\,\cdot\,, a)$ is the generalized singular value function.

Lemma 4.4

For every $\lambda > 0$,

$$\tau(1 - q^{(\lambda)}) \le \left|\{t > 0 : A(t) \ge \lambda\}\right|.$$

For $r = 1 - q_N^{(\lambda)}$, iteration of the corner estimate gives

$$\lambda^{\tau(r)} \le \Delta_r(rar) \le \exp\left(\int_0^{\tau(r)} \log \mu(s,a)\,ds\right).$$

This implies $\lambda \le A(\tau(r))$, and A is decreasing.

Summing the projection tails

Carleman's inequality gives

$$\int_0^\infty \exp\left(\frac{1}{t}\int_0^t \log f(s)\,ds\right)\,dt \le e\int_0^\infty f(t)\,dt.$$

Applying it to $f(t) = \mu(t,a)^p$ yields

$$\int_0^\infty A(t)^p\,dt \le e\,\|a\|_p^p.$$

Lemma 4.5

For a positive majorant $x_n \le a$ and $0 < p \le 1$,

$$\sum_{k\in\mathbb{Z}} 2^{kp}\tau(1-q^{(2^k)}) \le \frac{2^p e}{2^p-1}\,\|a\|_p^p.$$

These determinant and summation estimates do not require regularity.

The regularity estimate

The additional input is a noncommutative form of Gundy's inequality.

Lemma 5.2

For the Cuculescu projections of a positive martingale associated with a regular filtration,

$$\tau\big((1-q^{(\lambda)})x\big) \le 2c_{\mathrm{reg}}\,\lambda\,\tau(1-q^{(\lambda)}).$$

This estimate replaces a trace involving the terminal value x by the size of the stopping projection.

This is where regularity enters the square function argument.

The column–row decomposition

Construct increasing projections

$$e_{j,n} = \bigwedge_{k \geq j} q_n^{(2^k)}, \qquad \pi_{0,n} = e_{0,n}, \quad \pi_{j,n} = e_{j,n} - e_{j-1,n} \ (j \geq 1).$$

For each n, the $\pi_{j,n}$ are mutually orthogonal and sum to 1.

Triangular truncations at the preceding filtration level give

$$d_n x^c = \sum_{i \leq j} \pi_{i,n-1} \, d_n x \, \pi_{j,n-1},$$

$$d_n x^r = \sum_{i > j} \pi_{i,n-1} \, d_n x \, \pi_{j,n-1}.$$

Since the projections are in $\mathcal{M}_{n-1}$, both are martingale difference sequences, and

$$x = x^c + x^r.$$

Square function distribution estimates

Write

$$B_k = \tau(1 - q^{(2^k)}), \qquad Q_\ell = \left\| q^{(2^\ell)} x q^{(2^\ell)} \right\|_2^2,$$

and $\mathrm{d}(s, b) = \tau(\chi_{(s,\infty)}(|b|))$.

Lemma 5.3

For a positive $x \in L_2(\mathcal{M})$ and a regular filtration,

$$\mathrm{d}(2^\ell, S_c(x^c)) \le C_{\mathrm{reg}} \left(2^{-2\ell} Q_\ell + \sum_{k \ge \ell} B_k \right),$$

$$\mathrm{d}(2^\ell, S_r(x^r)) \le C_{\mathrm{reg}} \left(2^{-2\ell} Q_\ell + \sum_{k \ge \ell} B_k \right).$$

The bound separates a compressed L_2 term from a projection tail.

Completion of the H_p estimate

For a positive martingale, choose a positive majorant $x_n \le a$. Proposition 4.6 gives

$$\sum_{\ell\in\mathbb{Z}} 2^{\ell(p-2)} Q_\ell + \sum_{\ell\in\mathbb{Z}} 2^{\ell p} \sum_{k\ge\ell} B_k \le C_p \|a\|_p^p .$$

The dyadic distribution formula therefore yields

$$\begin{aligned} \|S_c(x^c)\|_p^p &\lesssim_p \sum_{\ell\in\mathbb{Z}} 2^{\ell p} \mathrm{d}(2^\ell, S_c(x^c)) \\ &\le C_{p,\mathrm{reg}} \|a\|_p^p . \end{aligned}$$

The same bound holds for $S_r(x^r)$. Taking the infimum over majorants,

$$\|x\|_{H_p} \le C_{p,\mathrm{reg}} \|(x_n)_n\|_{L_p(\mathcal{M},\ell_\infty)} .$$

Positive decomposition gives the extension to general martingales.

Operator-valued Hardy spaces

Let $(\mathcal{N}, \varphi)$ be semifinite and set

$$\mathcal{M} = L_\infty(\mathbb{R}) \bar{\otimes} \mathcal{N}.$$

For the Poisson extension $P_\varepsilon f$, define

$$G_c(f)(s) = \left(\int_0^\infty \varepsilon \left| \partial_\varepsilon P_\varepsilon f(s) \right|^2 d\varepsilon \right)^{1/2}.$$

Mei's spaces satisfy

$$\|f\|_{\mathcal{H}_1^c} = \|G_c(f)\|_1, \qquad \mathcal{H}_1 = \mathcal{H}_1^c + \mathcal{H}_1^r, \qquad \|f\|_{\mathcal{H}_1^r} = \|f^*\|_{\mathcal{H}_1^c}.$$

For $h_1, h_2 > 0$, consider the interval averages

$$A_h f(t) = \frac{1}{h_1 + h_2} \int_{t-h_1}^{t+h_2} f(s)\, ds.$$

The operator-valued Hardy estimate

Theorem 6.3 (Operator-valued Hardy estimate)

For every $f \in L_1(\mathcal{M})$,

$$\|f\|_{\mathcal{H}_1(\mathbb{R},\mathcal{N})} \leq C \left\| (A_h f)_{h \in \mathbb{R}_+^2} \right\|_{L_1(\mathcal{M},\ell_\infty)},$$

where C is universal.

Use the dyadic filtration, whose regularity constant is 2:

$$\begin{aligned} \|f\|_{\mathcal{H}_1(\mathbb{R},\mathcal{N})} &\lesssim \|f\|_{H_1(\mathcal{M},\mathcal{D})} \\ &\lesssim \|(\mathbb{E}_n f)_n\|_{L_1(\mathcal{M},\ell_\infty)} \\ &\lesssim \|(A_h f)_h\|_{L_1(\mathcal{M},\ell_\infty)}. \end{aligned}$$

The first step is Mei's Hardy space comparison. The second step is our Davis inequality.

Noncommutative martingale BMO

For a noncommutative probability space, define

$$\|f\|_{\mathcal{BMO}^c} = \sup_{n\geq 1} \left\| \mathcal{E}_n\left(|f-\mathcal{E}_{n-1}f|^2\right)\right\|_\infty^{1/2}, \qquad \mathcal{E}_0 f = 0,$$

and

$$\|f\|_{\mathcal{BMO}^r} = \|f^*\|_{\mathcal{BMO}^c}.$$

The two-sided space is

$$\mathcal{BMO}(\mathcal{M}) = \mathcal{BMO}^c(\mathcal{M}) \cap \mathcal{BMO}^r(\mathcal{M}),$$

with

$$\|f\|_{\mathcal{BMO}} = \max\{\|f\|_{\mathcal{BMO}^c}, \|f\|_{\mathcal{BMO}^r}\}.$$

The duality $H_1(\mathcal{M})^* = \mathcal{BMO}(\mathcal{M})$ connects the Davis inequality with positive BMO decompositions.

A positive BMO decomposition

Theorem 6.6

Assume that the filtration is regular. For every $f \in \mathcal{BMO}(\mathcal{M})_+$, there exist $g \in \mathcal{M}_+$ and $u_k \in \mathcal{M}_+$ such that

$$f = g + \sum_{k \geq 1} \mathcal{E}_k(u_k) \qquad \text{in the strong operator topology,}$$

and

$$\|g\|_\infty + \left\| \sum_{k \geq 1} u_k \right\|_\infty \leq C_{\mathrm{reg}} \|f\|_{\mathcal{BMO}} .$$

Talebi, Moslehian and Sadeghi (2019) obtained this representation assuming the endpoint Hardy–maximal estimate.

Our theorem supplies that estimate for regular filtrations.

Dyadic Carleson decomposition

Let $\mathcal{M} = L_\infty(0,1)\bar{\otimes}\mathcal{N}$ with the dyadic filtration, where $(\mathcal{N}, \varphi)$ is a noncommutative probability space.

Theorem 6.7

For every $f \in \mathcal{BMO}(\mathcal{M})_+$, there exist $g \in \mathcal{M}_+$, dyadic intervals I_i, and $a_i \in \mathcal{N}_+$ such that

$$f = g + \sum_i a_i \chi_{I_i}, \qquad \|g\|_\infty \le \|f\|_{\mathcal{BMO}},$$

and, for every dyadic interval I_0,

$$\left\| \frac{1}{|I_0|} \sum_{i:\, I_i \subset I_0} a_i |I_i| \right\|_{\mathcal{N}} \le C \|f\|_{\mathcal{BMO}}.$$

Here C is universal.

This is the operator-valued counterpart of the dyadic Garnett–Jones decomposition.

Selected references

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Thank you!